

Surya Vamsi Gara

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Education

VIT-AP University

Bachelor of Technology in Computer Science, 8.05

2023 - 2027

Amaravati, AP

Professional Summary

AI developer with hands-on experience building agentic systems from prototype to production. Built knowledge graph extraction pipelines, hybrid RAG systems, autonomous coding agents, and deployed services on AWS with Docker.

Projects

GraphSleuth - Document Intelligence & Knowledge Graph Platform | Python, FastAPI, Supabase, React.js [GitHub](#) | [Live](#)

- Built an intelligence tool that ingests unstructured documents and extracts structured entity-relationship knowledge graphs via LLM-driven extraction, turning raw text into queryable, traversable data assets.
- Engineered a three-stage entity deduplication pipeline (alias matching → token overlap → embedding cosine similarity) to consolidate fragmented extractions across document chunks.
- Implemented guided beam-search retrieval that scores graph neighbors against query embeddings, bounding traversal cost while surfacing the most relevant paths enabling efficient querying of dense knowledge networks.
- Containerized with Docker and deployed on AWS EC2 with nginx reverse proxy, delivering a production-grade API for real-time document Q&A with full reasoning traceability

Tenet (Terminal-Based AI Coding Agent) | Python, OpenAI API

[GitHub](#)

- Architected a production-grade AI agent with a Reason, Act, Observe loop using OpenAI API, implementing multi-step autonomous workflows that explore codebases, plan multi-file changes, seek explicit user approval, and verify results.
- Designed two-tier memory system: persistent Project Context (pinned architectural facts, file summaries, read-file tracking) plus a trimmable Conversation Window with safe-boundary compression, solving the context retention challenge in long-running agent sessions.
- Built a **safety layer** for destructive operations - path traversal rejection, mandatory confirmation for delete_path and destructive shell patterns (rm, DROP TABLE,...).
- Packaged as a global CLI tool via uv tool install, following the git-style convention of operating on whichever directory the command is run from.

AskRepo - Full Stack Codebase AI Assistant | Python, FastAPI, PostgreSQL, AWS, React.js

[GitHub](#) | [Live](#)

- Architected an end-to-end full-stack RAG application enabling real-time streaming chat interactions with entire public GitHub repositories.
- Developed a hybrid semantic search backend combining vector embeddings (pgvector) and keyword search (BM25) via Reciprocal Rank Fusion (RRF), improving context extraction accuracy by 40%
- Implemented an Abstract Syntax Tree (AST) parsing engine using tree-sitter to intelligently chunk Python and JavaScript code by logic blocks (classes/functions).
- Built an asynchronous data pipeline using Celery and Redis to manage repository cloning and high-volume embedding generation without API blocking.
- Containerized and deployed services via Docker Compose on AWS EC2.

Technical Skills

AI: Agentic Workflows, RAG Pipelines, Vector Databases, Hybrid Search, Model Context Protocol (MCP), PyTorch, NumPy

Backend: Python, FastAPI, AsyncIO, RESTful APIs, PostgreSQL

Frontend: React.js, JavaScript

DevOps & Infrastructure: Docker & Docker Compose, Linux/WSL, Git